RISA

Product Designer Assignment

ClarioRx

AI-Powered Agentic Browser for Prior Authorization

> “ Transforming Prior Authorization Through Agentic Automation ”

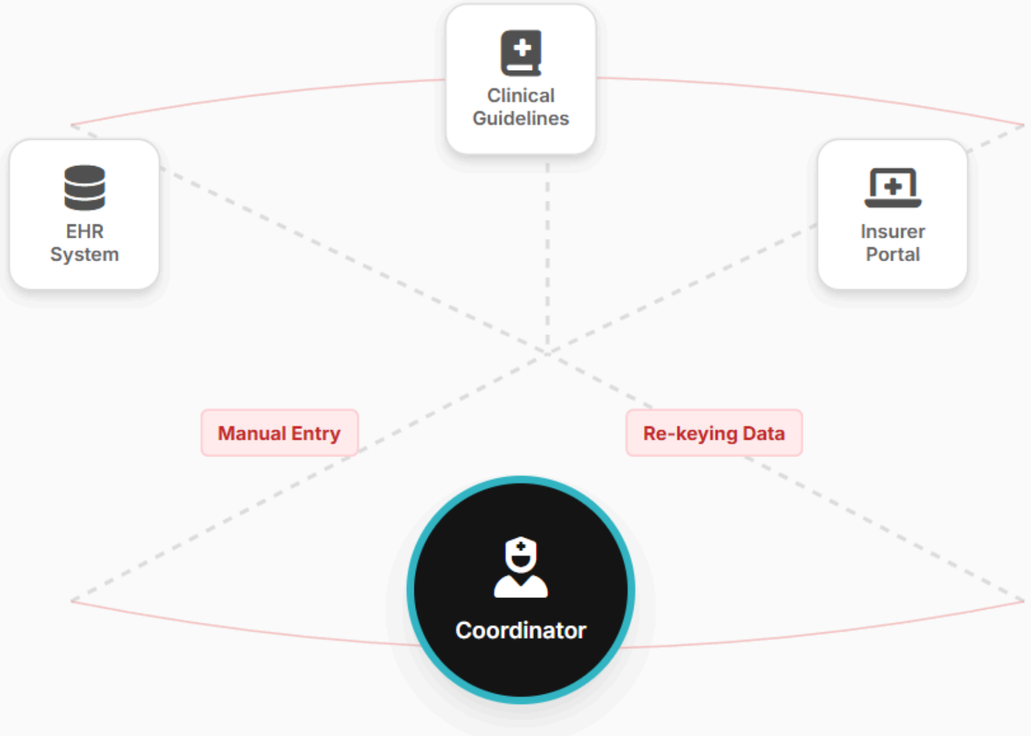
Saswat Dash

21ME31067

CONTEXT & PROBLEM DEFINITION

Prior Authorization is Fragmented & Manual

The current workflow relies on human coordinators to bridge the gap between incompatible technologies.



Operational Bottlenecks

- **Manual Extraction:** Clinical data is trapped in notes and must be found manually.
- **Multiple Systems:** Staff log into 20+ different payer portals with unique UIs.
- **Inconsistent Requirements:** Rules change frequently without notification.

PROCESSING TIME

30–45 min

Per single authorization request

PATIENT DELAY

2–3 weeks

Average time to schedule procedure

UNDERSTANDING THE USER

User Reality & Workflow Impact

Sarah Jenkins

AUTHORIZATION COORDINATOR

RCM Specialist • 6 Years Exp.

DAILY VOLUME	PORTALS USED
20–40 cases	**15+** unique logins

> *"I spend 80% of my day just hunting for clinical notes in the EHR and pasting them into different insurer portals. It feels like data entry, not healthcare."*

PRIMARY GOALS

Faster Approvals · Clear Status Visibility · Zero Denials

TOP FRUSTRATIONS

- Scattered clinical requirements
- Portal timeouts & re-work
- Endless phone/fax follow-up loops

STORY 1 The Volume Bottleneck

Routine MRI Authorization Request

28	5	2
REQUESTS QUEUED	COMPLETED BY 5PM	RETURNED ERRORS

→ **Outcome:** Patient procedure delayed by 4 days due to administrative backlog.

STORY 2 Critical Care Blockage

Urgent Cardiac Catheterization

- Coordinator uploads 45-page PDF but misses the "Stress Echo" report buried in a separate file.
- Insurer algorithm auto-denies for "Insufficient Clinical Evidence".

Patient Hospitalized

While waiting for appeal

Key Insight: Delays are not abstract—they directly impact care quality and patient outcomes.

CURRENT STATE ANALYSIS

Workflow Breakdown

Identify
Check eligibility & requirements

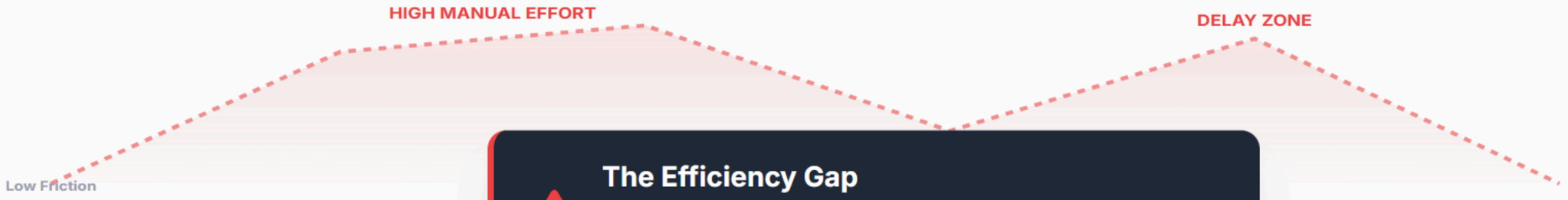
Gather Data
Hunt for clinical notes in EHR

Fill Forms
Manual re-keying into portal

Submit
Upload docs & hit send

Follow-up
Phone tag & status checks

Approve
Final authorization received



The Efficiency Gap

Time, error rate, and frustration peak in the **Data Gathering**, **Form Filling**, and **Follow-up** stages.

SOLUTION ARCHITECTURE

ClarioRx Agentic Browser

Transform prior authorization into an automated, agent-assisted browser workflow with **minimal human intervention**.

INPUT SOURCE 1
EHR Data

INPUT SOURCE 2
Payer Rules

AI Agent Automation

Intelligent browser overlay that reads, interprets, and fills portal requirements autonomously.

Visual Perception

Clinical Logic

Auto-Execution

DESIGN PHILOSOPHY

Core Design Principles

Guiding the interaction between clinical expertise and agentic automation.

Persistent Context

Critical patient data remains visible across all navigation states, reducing cognitive load and memory reliance.

Transparency & Traceability

Every agent action is logged, sourced to clinical documentation, and easily verifiable by human reviewers.

Automation First

Defaults to agent execution for data entry and retrieval, elevating the human role to validation and judgment.

Progressive Disclosure

Complex insurer rules are hidden by default, revealing specific requirements only when relevant to the case.

Visual Consistency

Standardizes diverse and chaotic payer portal interfaces into a unified, predictable interaction model.

SCREEN 1

VISUAL DESIGN CHOICES

Color System: Meaningful, Not Decorative

Purpose: Drive attention and decision-making

- **Teal (#32B8C6) → Primary action + AI activity**

  Communicates automation, progress, and momentum
- **Red / Yellow status tags → Urgency & risk signaling**

  Enables instant prioritization without reading text
- **Neutral surfaces (#F8F9FA, #FFFFFF) → Low-contrast workspace**

  Reduces eye fatigue during long workflows
- **Soft accent backgrounds (5% washes) → Highlight active elements subtly**

  Avoids visual noise while still guiding focus

Outcome: Users instantly know
"What requires attention?" and "What the agent is doing?"

Typography: Hierarchy That Mirrors Workflow

- **Font Choices: Inter + JetBrains Mono**
  - Inter for readability in dense clinical text

    → clean shapes, optimized for UI screens
  - JetBrains Mono for codes (CPT, ICD)

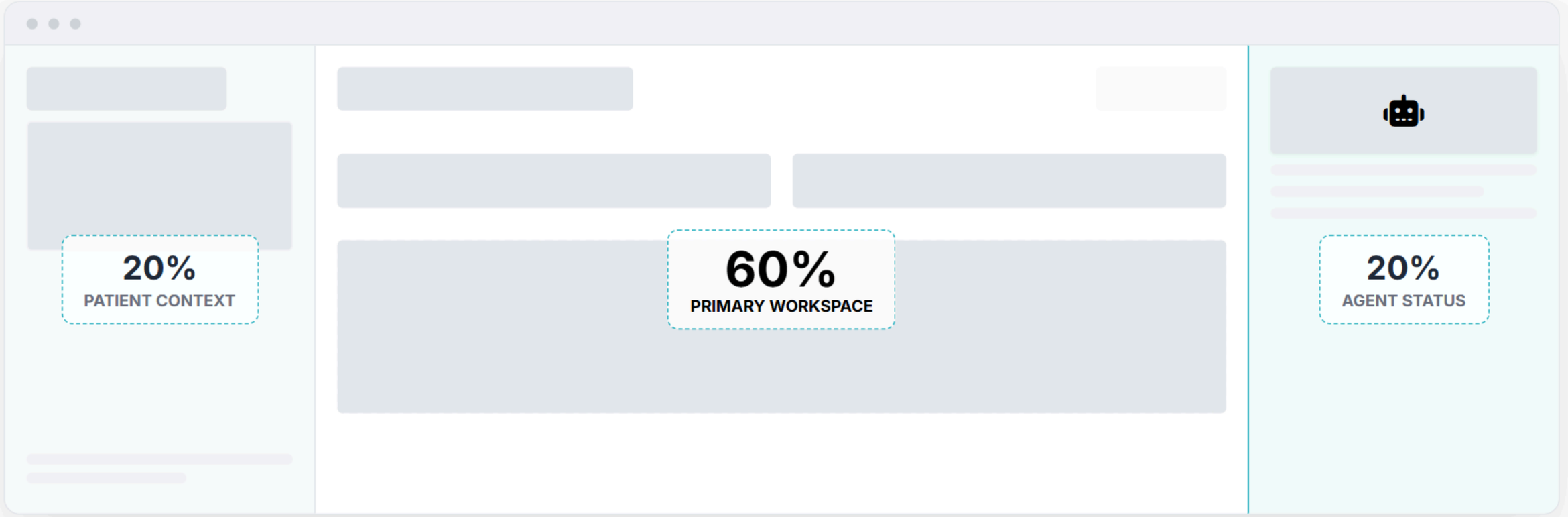
    → reinforces precision and medical coding context
- **Hierarchy Rules**
  - Larger, bold: patient & task headers
  - Medium: actionable content (procedures, CTAs)
  - Small, muted: metadata & timestamps

Outcome:
Users consume information in the correct order without effort

STRUCTURAL RATIONALE

20/60/20 Layout Architecture

Optimizing screen real estate for cognitive safety and automation oversight.



Persistent Context

Patient data remains visible, preventing memory errors during task switching.

Focused Efficiency

Central workspace maximizes screen real estate for the active task.

Automation Visibility

Users can monitor agent activity peripherally without distraction.

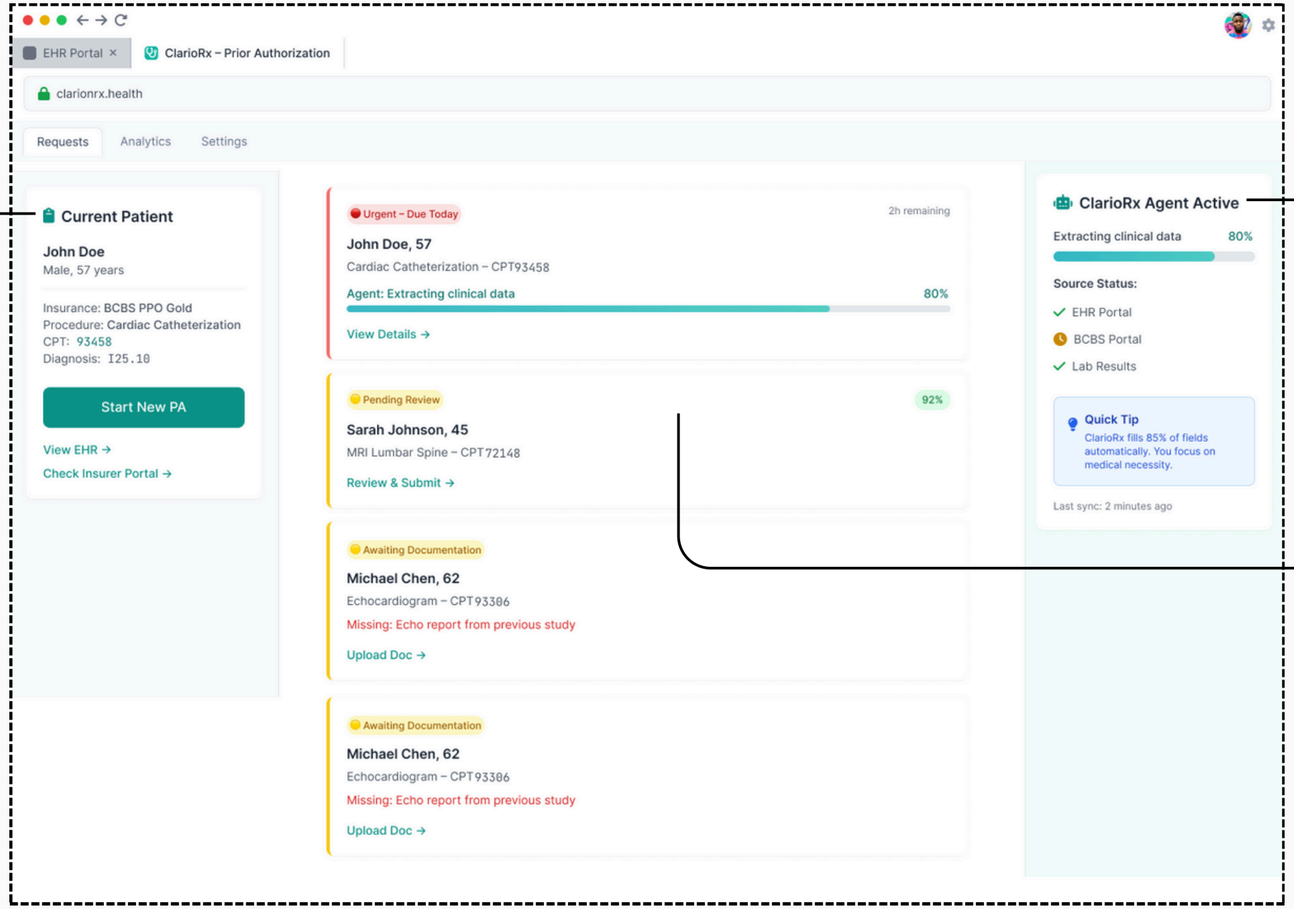
Low Cognitive Load

Predictable zoning reduces the mental effort of finding information.

SCREEN 1

DASHBOARD

AI-Driven Workflow Control Center



Left – Patient Context (Safety Anchor)

- "Current Patient" card with demographics, insurance, procedure and diagnosis.
- Primary CTA: Start New PA
- Secondary links: View EHR →, Check Insurer Portal →
- Purpose: keep WHO + WHAT visible at all times so the coordinator never loses track of the active patient/request. This is critical in healthcare to avoid wrong-patient errors.

Right Panel — Agent Activity & Trust Signals

- Shows the AI actively extracting clinical data (80% progress)
- Displays connected data sources with status icons
- Includes educational "Quick Tip"
- Last sync timestamp reinforces system reliability

Center Panel — Prioritized Request Queue

- Stacked cards representing active authorization requests
- Urgency states: Urgent, Pending Review, Awaiting Documentation
- Real-time agent progress bar on the top request
- Clear CTAs: View Details, Review & Submit, Upload Doc
- Rationale:
- Visual hierarchy directs attention to highest-value tasks, reinforcing efficiency and reducing decision fatigue.

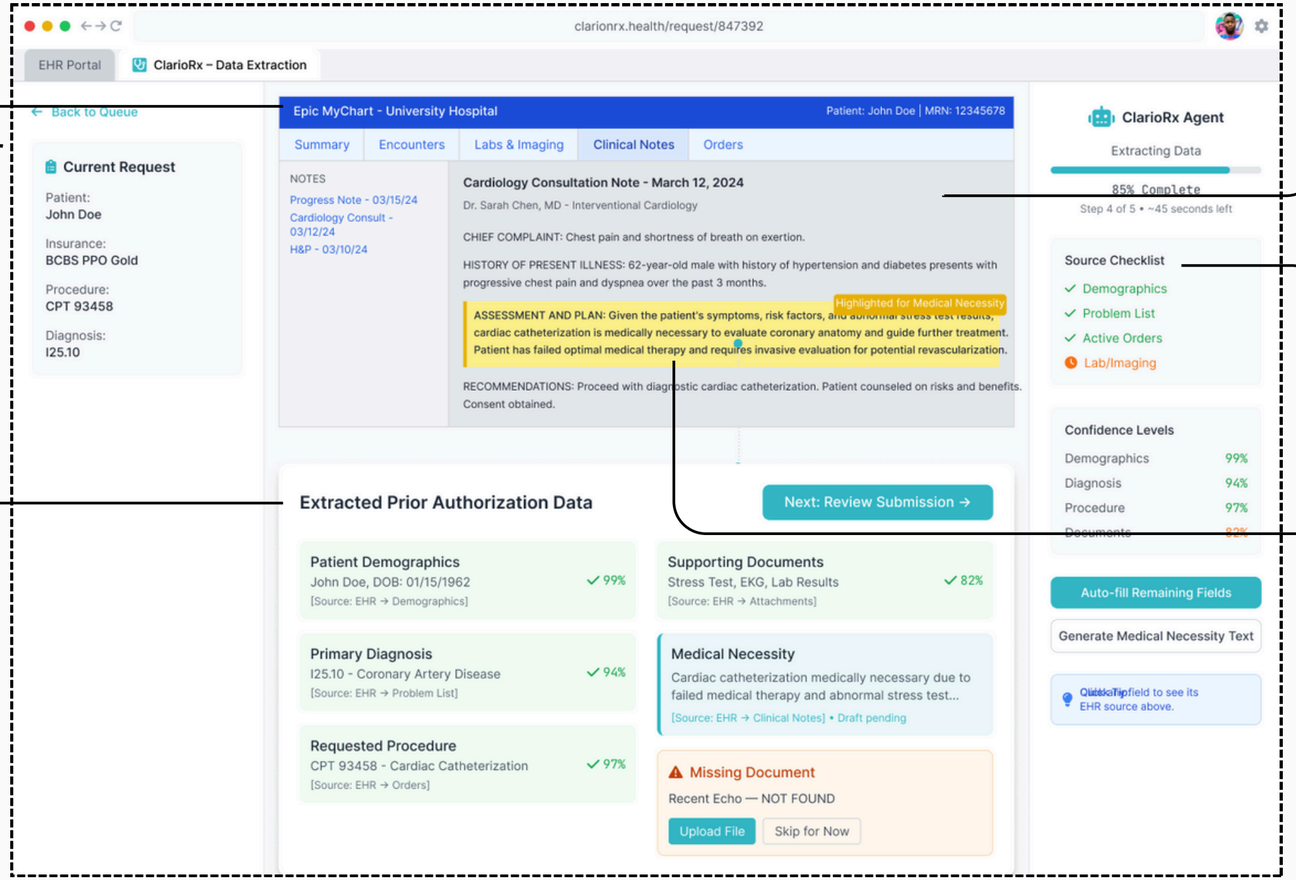
Why This Screen Is the Hero Moment ?

It proves the core value proposition instantly:
"The browser is doing the administrative work for you so you can focus on clinical judgment."
Evaluators see automation, prioritization, and safety — all within three seconds.

SCREEN 2

DATA EXTRACTION

Automated Data Extraction & Structuring



CENTER PANEL – The Transformation Layer

- Top Section: EHR Live View
- Embedded EHR content
- Tabs mimicking Epic UX (Summary / Encounters / Labs...)
- Highlighted clinical note section
- This establishes:
  - Real-world data source
  - Unstructured input format
  - Medical justification location
- The yellow highlight is GOLD here.
- It visually proves:
- “The AI is reading THIS text and extracting meaning.”
- This is the transparency most AI products forget.

Bottom Section: Extracted Prior Authorization Data

- This is the payoff.:
- Clean structured fields
- Confidence percentages
- Source traceability labels
- Missing document state
- Drafted Medical Necessity
- This bottom section tells reviewers:

“The AI didn't hallucinate. It mapped specific values to specific PA fields from specific sources.”

Live EHR Data Source

- The agent reads directly from the original clinical record, just as a human would.
- Raw notes, reports, and imaging references are processed in real time without manual copy-paste.

Confidence & Source Mapping

- Each field shows extraction confidence and its exact data source.
- Users can verify accuracy instantly and trace every value back to its originating EHR section.

AI Highlighted Medical Necessity Text

- Relevant justification sentences are automatically identified and highlighted.
- This makes the clinical basis for the procedure visible, traceable, and audit-ready.

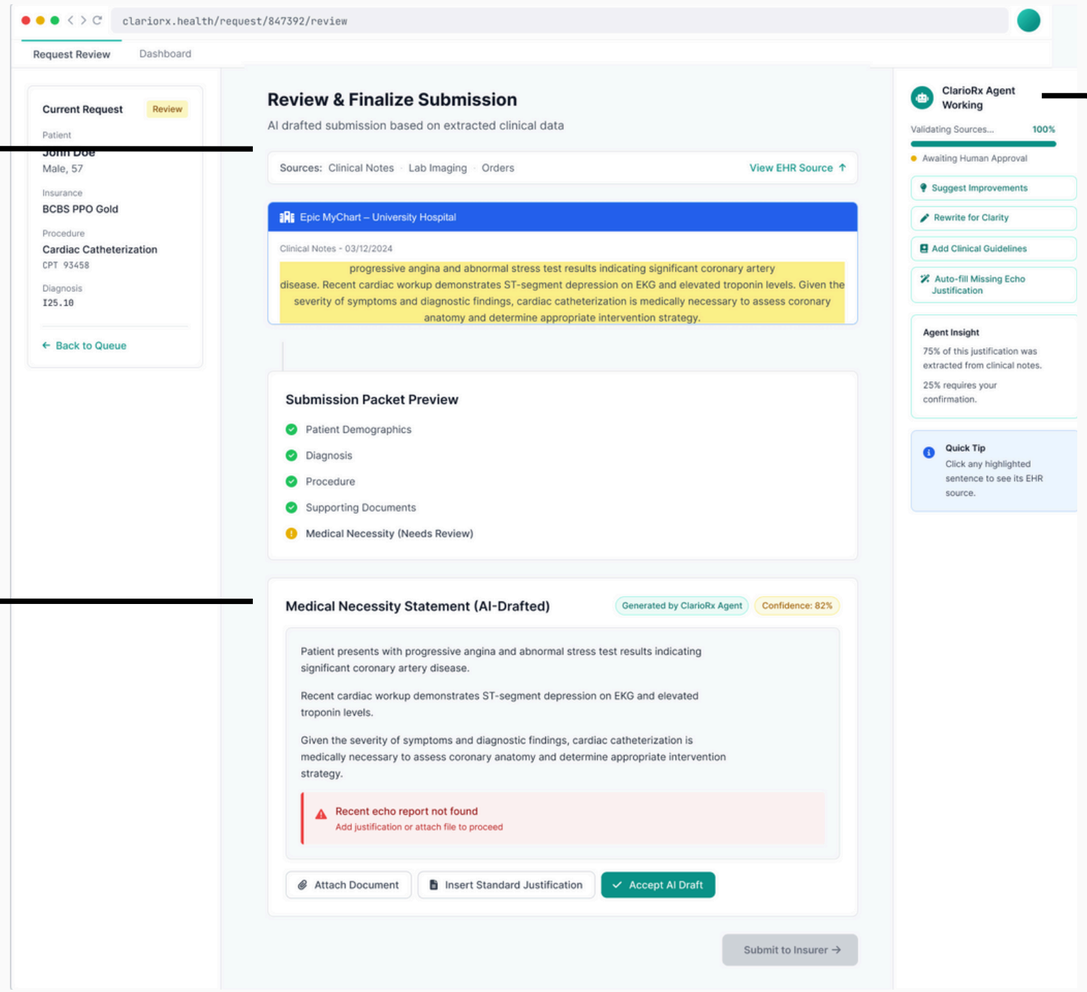
Interaction & Workflow Logic

1. Agent extracts from EHR (auto)
2. Highlights relevant medical necessity text
3. Populates PA fields
4. Flags missing documentation
5. Prepares draft justification

SCREEN 3

REVIEW & FINALIZE

Human Validation & Clinical Oversight



Center Panel — Review Workspace

This panel contains the two core review zones:

- Submission Packet Preview
- Checklist showing completion state by category
  - Green items: auto-complete and verified
  - Yellow: requires human review (Medical Necessity)

Effect:

Instant understanding of what remains before submission.

Medical Necessity Statement (AI-Drafted)

- Agent-generated justification text based on clinical notes
- Confidence badge displayed
- Highlighted missing echo report
- Clear CTA options:
- Attach Document
- Insert Standard Justification
- Accept AI Draft

Rationale:

This is the only step where clinical judgment is required. The interface isolates it and minimizes cognitive load.

Right Panel — Agent Assistance & Guidance

- Displays completion status: Validating Sources → Awaiting Human Approval
- Shows AI contribution percentage
- Offers smart action options:
- Suggest Improvements
- Rewrite for Clarity
- Add Clinical Guidelines
- Auto-fill Missing Echo Justification

Rationale:

The system supports the user in completing the final 15–25% that automation cannot finalize reliably.

Core Interaction Principles Demonstrated

Human-in-the-loop validation
The AI drafts the justification, but final approval remains with the clinician.

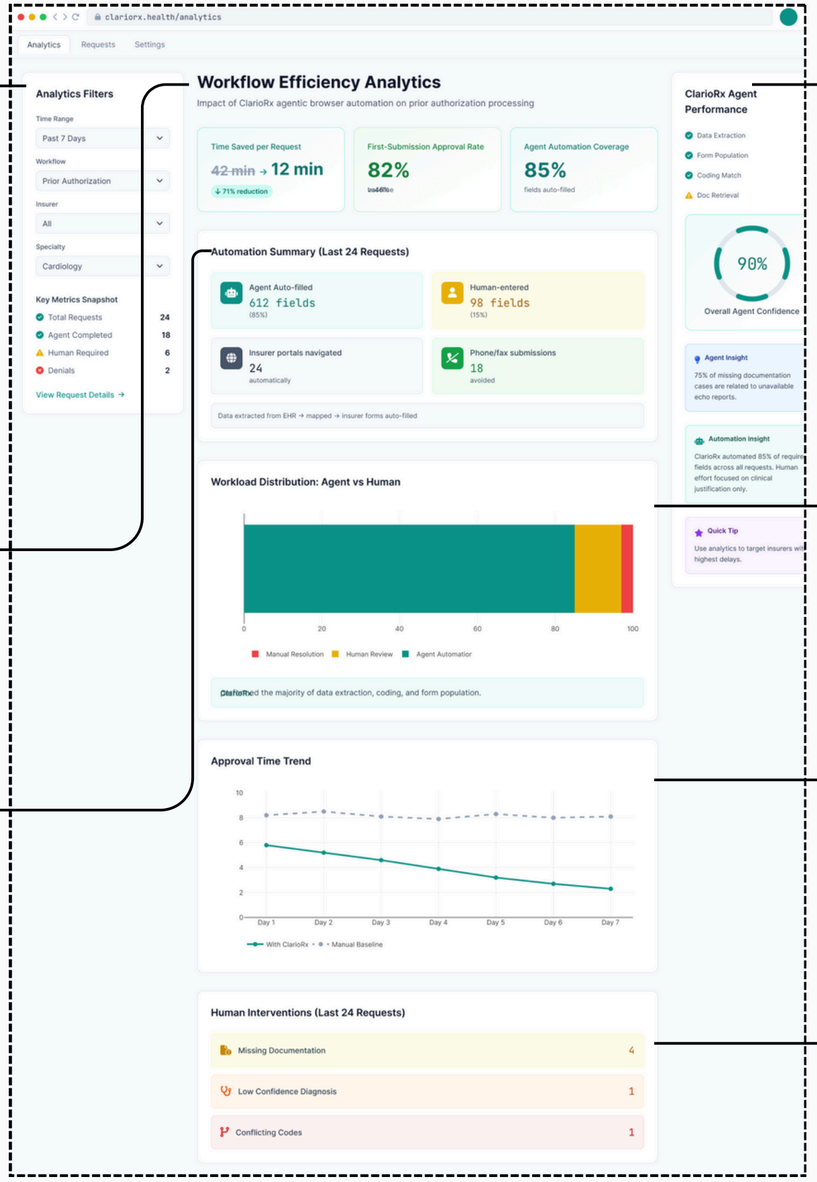
Transparency of automation contribution
Agent insight clearly states what was extracted automatically vs. what needs human sign-off.

Focused human effort
Only one blocking item (missing echo) prevents submission

SCREEN 4

ANALYTICS

Workflow Efficiency & Automation Insights



Left Panel — Analytics Filters & Key Metrics

- Time range, workflow type, insurer, specialty selectors
- Key metric snapshot:
  - Total Requests
  - Agent Completed
  - Human Required
  - Denials
- Link to request details

Rationale:
Users can segment performance data quickly, enabling targeted insights such as insurer-specific delays or workflow bottlenecks.

Center Panel — Performance Dashboards

- This section communicates outcome value clearly through:
- Efficiency Metrics
  - Time saved per request (42 min → 12 min)
  - First-submission approval rate improvement
  - Automation coverage (85% of fields auto-filled)

Effect:
Makes ROI of automation immediately visible.

Automation Summary (Last 24 Requests)

- Agent auto-filled fields
- Human-entered fields
- Insurer portals navigated automatically
- Phone/fax submissions avoided

Rationale:
Highlights real-world operational improvements and workload reduction.

Right Panel — Agent Performance & Insights

Category-level accuracy metrics:
Data Extraction
Form Population
Coding Match
Documentation Retrieval
Overall agent confidence (90%)
Insight callouts:
Documentation retrieval weakness
Automation strength areas
Optimization suggestions
Rationale:
Supports trust-building and provides actionable process intelligence.

Workload Distribution: Agent vs Human

- Horizontal bar representing automation vs manual work

Effect:
Visual proof that the agent performs the majority of data extraction and form population.

Approval Time Trend

- Line chart comparing automated vs manual baseline over 7 days

Rationale:
Shows sustained time reduction and consistency improvements.

Human Interventions Log

- Missing documentation
- Low confidence diagnosis
- Conflicting codes

Effect:
Identifies automation gaps and improvement opportunities.

MARKET POSITION & IMPACT

Differentiation & Outcomes

From fragmented manual entry to unified agentic intelligence.

CAPABILITY	Manual Process	Insurer Portals	EHR Modules	ClarioRx
Workflow Velocity	Slow	Med	Med	✓
Agentic Auto-Fill	×	×	Partial	✓
Clinical Traceability	×	×	✓	✓
Cross-Payer Unification	×	×	×	✓
Cognitive Safety	High Load	High Load	Med Load	Optimized

71%
TIME REDUCTION
↓ 42m → 12m

85%
AUTO-FILLED
Of all required fields

82%
FIRST-PASS APPROVAL
↑ vs Industry Avg

-18
CALLS / FAXES
Fewer interventions per case

24 Portals
FULLY AUTOMATED

Thank You

Link for Design: https://www.figma.com/design/TZIRnKSvCs1XFM41dvKstv/Risa-Lab?node-id=20-4214&t=ln36qrBr7Zj46JKd-1